Machine Learning Department
School of Computer Science
Carnegie Mellon University

Linear Regression

+

Optimization for ML

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Lecture 8
Feb. 4, 2026

Winter Olympics!

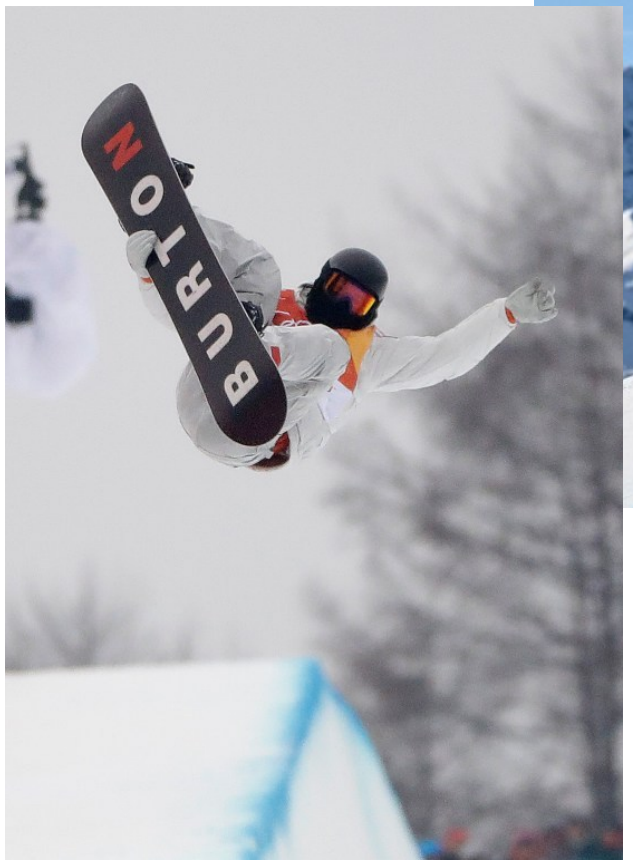


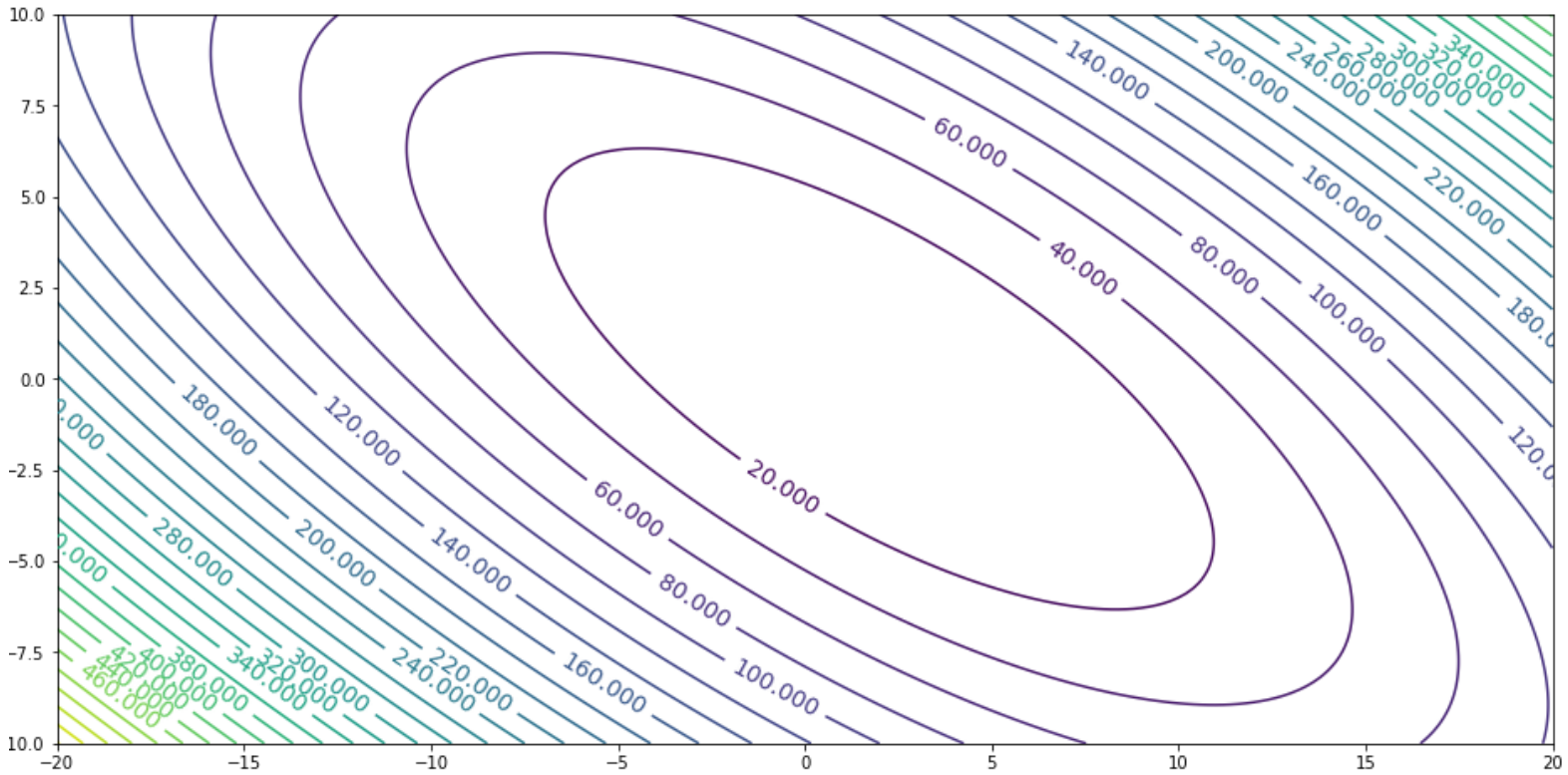
Image credit: AP (Gregory Bull), NBC News, USA Today, Altius

GRADIENT DESCENT:
LEARNING RATE HYPERPARAMETER

Polls 1 and 2

Linear regression gradient descent

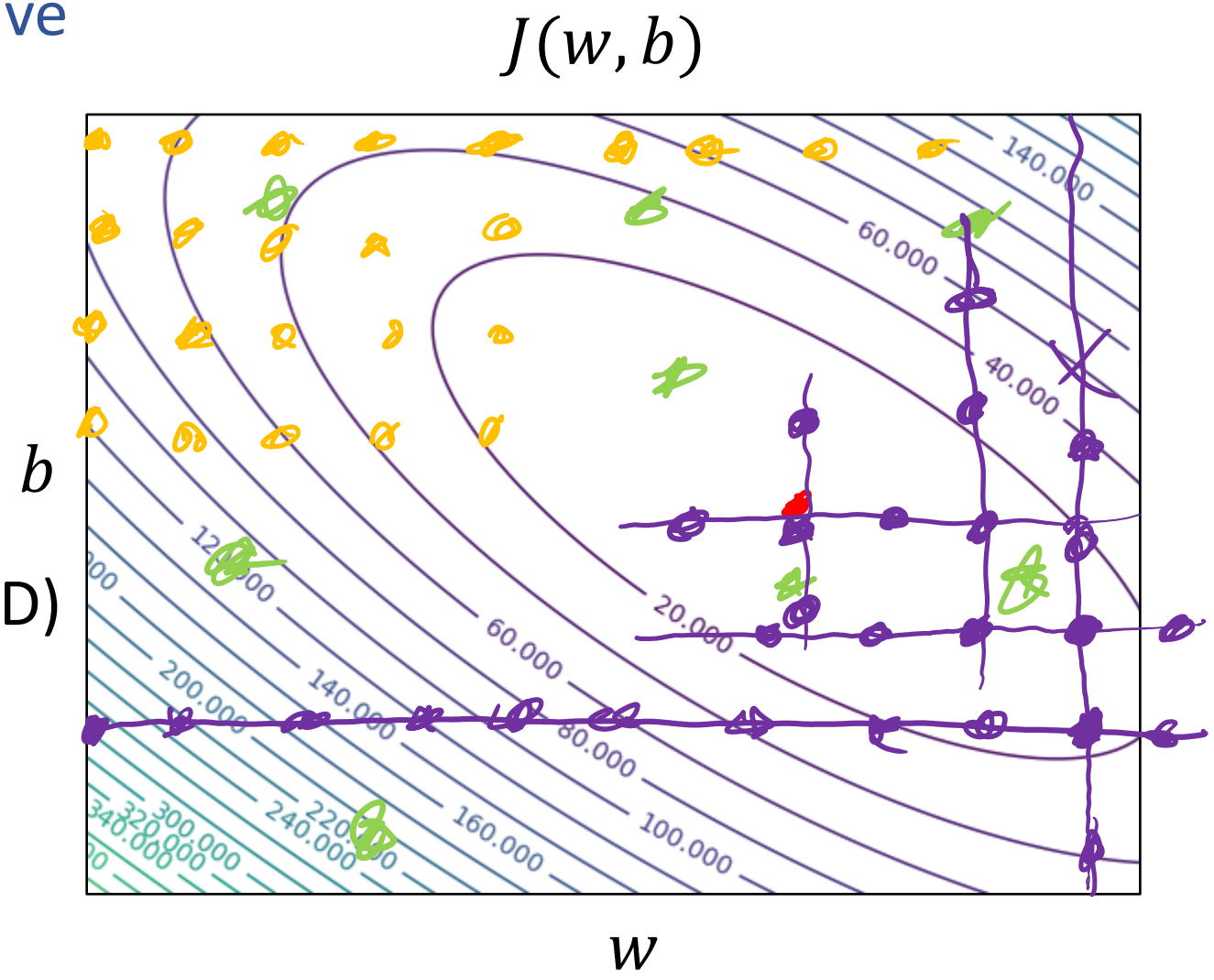
What are the axis labels for this plot?



Linear Regression

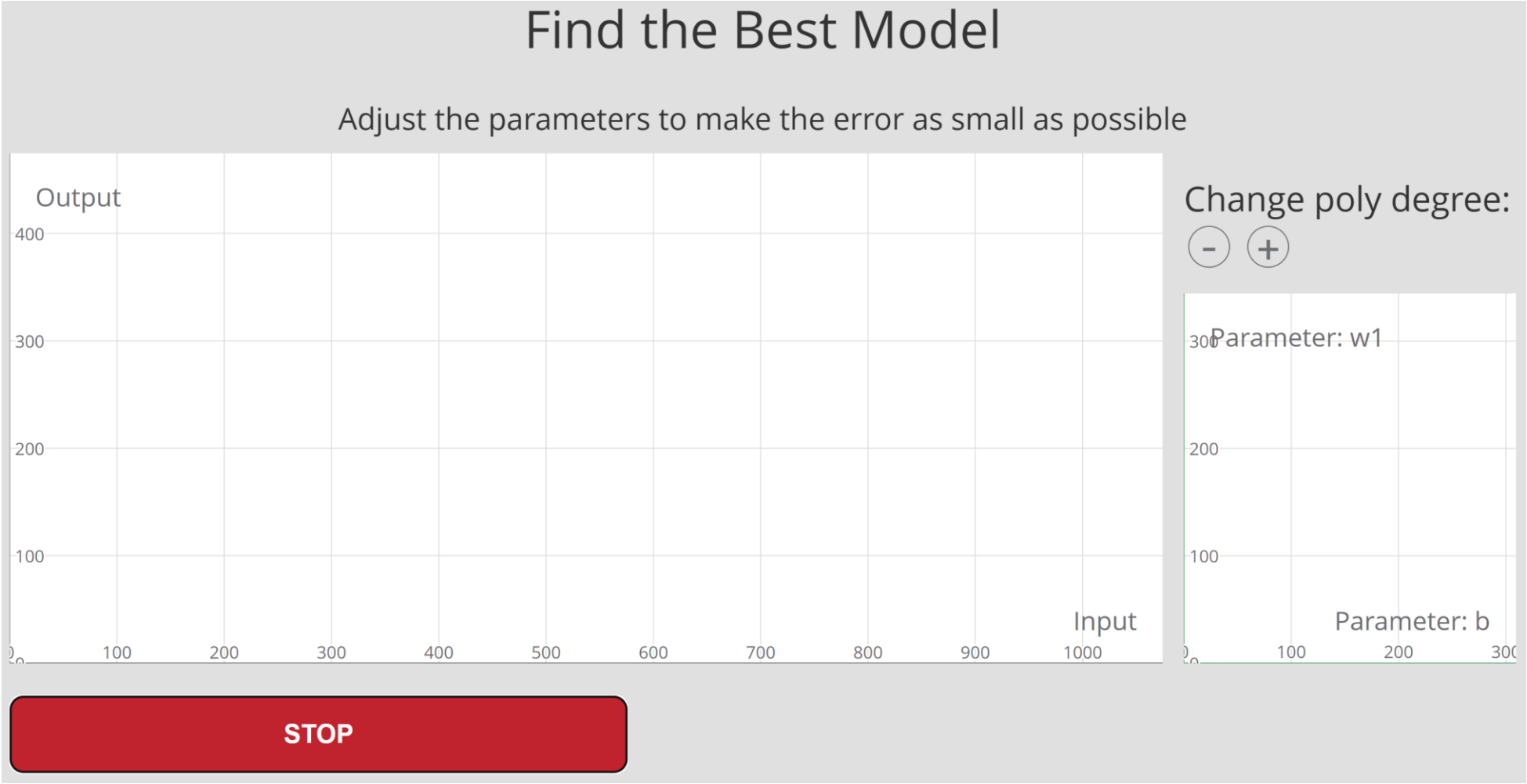
Methods for optimizing the objective

- Closed-form solution
- Grid search
- Random search
- Gradient descent
- Stochastic gradient descent (SGD)
- Minibatch SGD
- Coord descent (slider) grid



Demo: Linear Regression Optmization

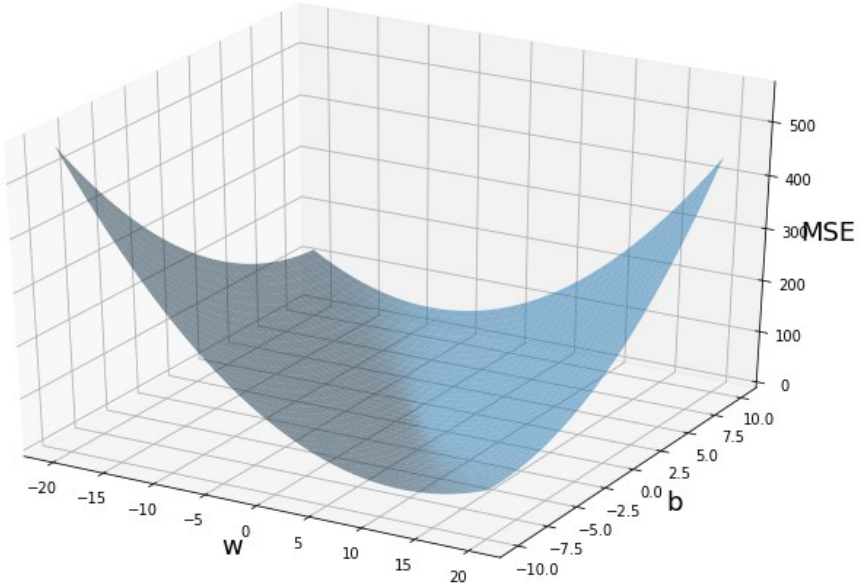
https://www.cs.cmu.edu/~pvirtue/AIS/regression/regression_poly.html

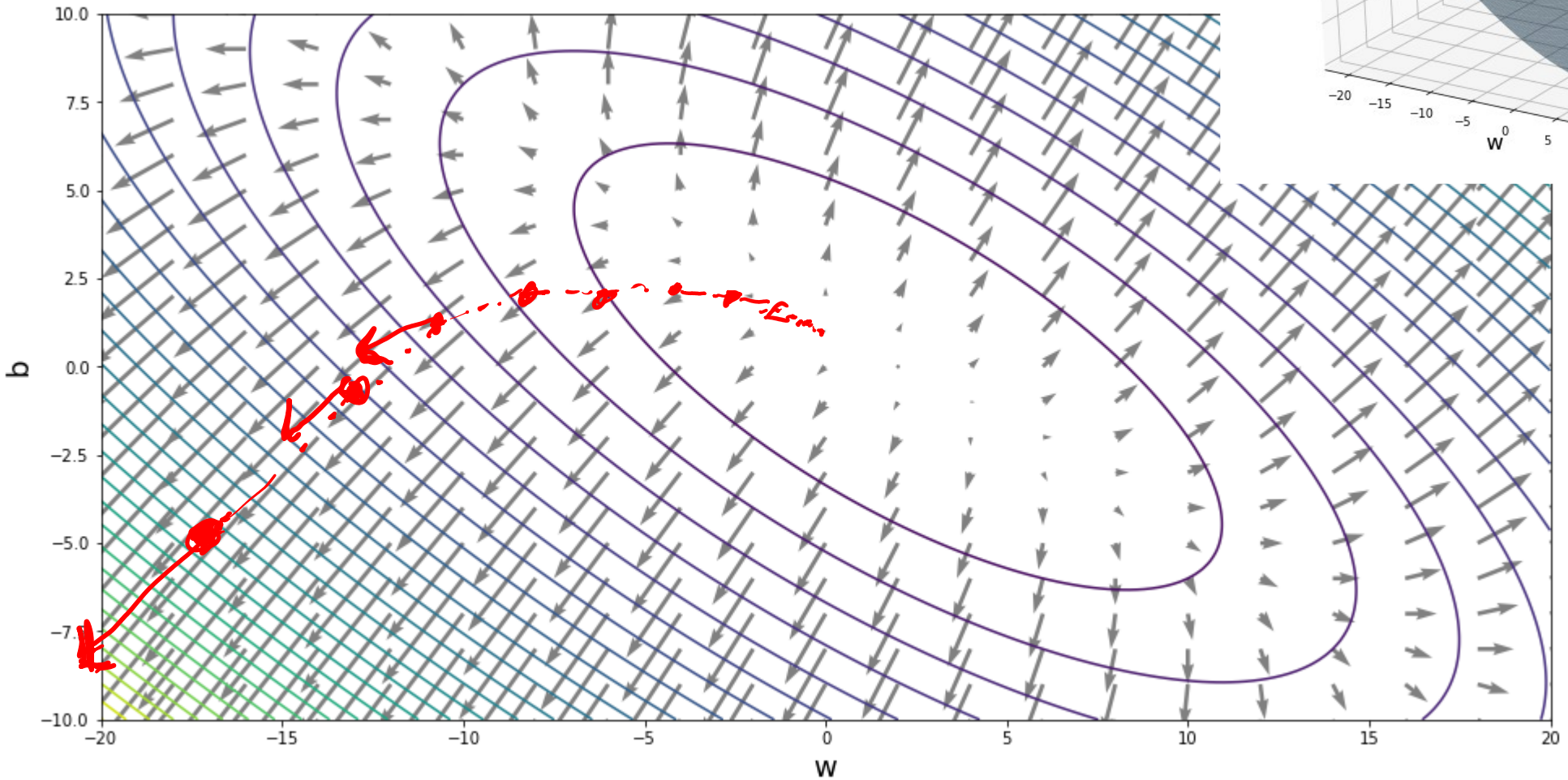


Optimization

Gradients

$$\nabla_\theta J = \frac{\partial J}{\partial \theta} = \begin{bmatrix} \partial J / \partial w \\ \partial J / \partial b \end{bmatrix}$$





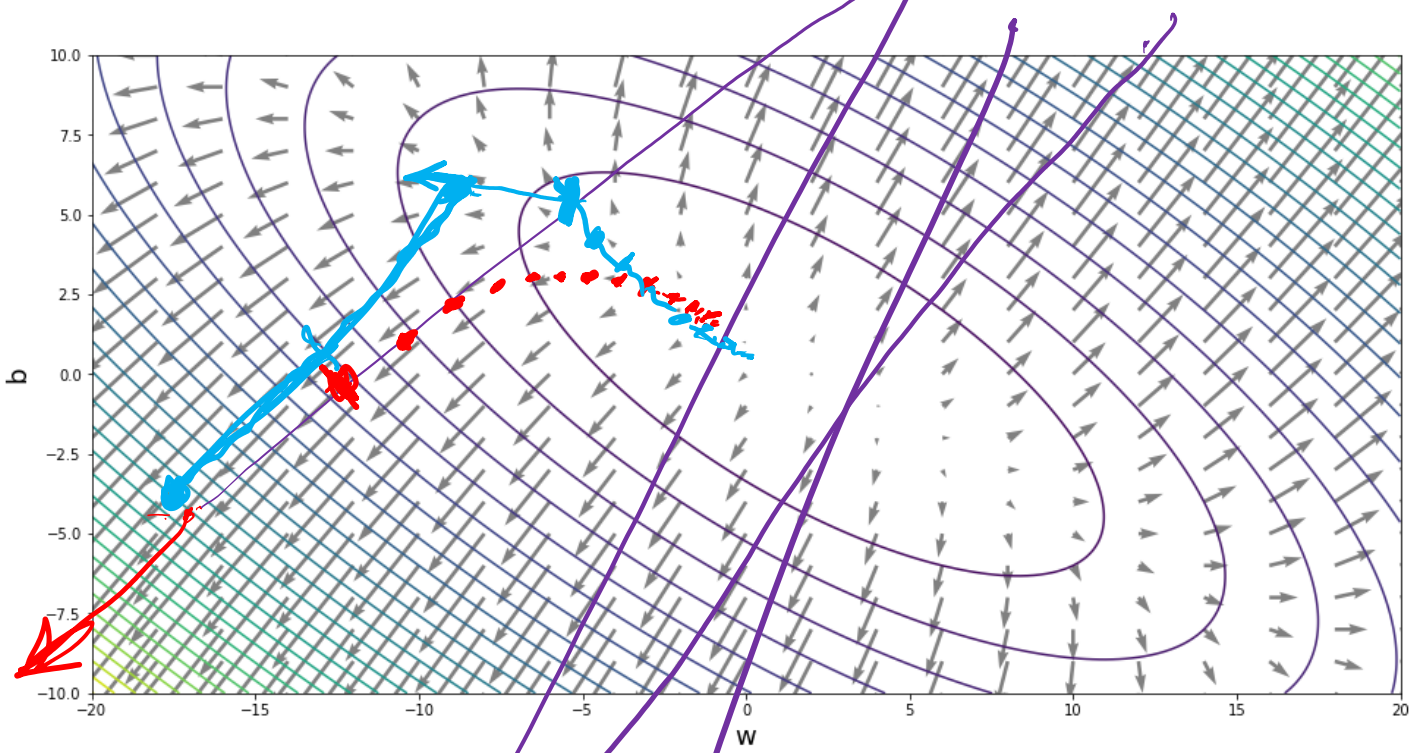
Optimization

Gradient descent

Hyperparameter:

learning rate γ (or sometimes α, η)

$\gamma = 10$
$\gamma = 2$
$\gamma = 1$



While not converged

Compute gradient: $\nabla J = \left[\frac{\partial J}{\partial w}, \frac{\partial J}{\partial b}\right]^{\top}$

Update parameters

$$w \leftarrow w - \gamma \frac{\partial J}{\partial w}$$

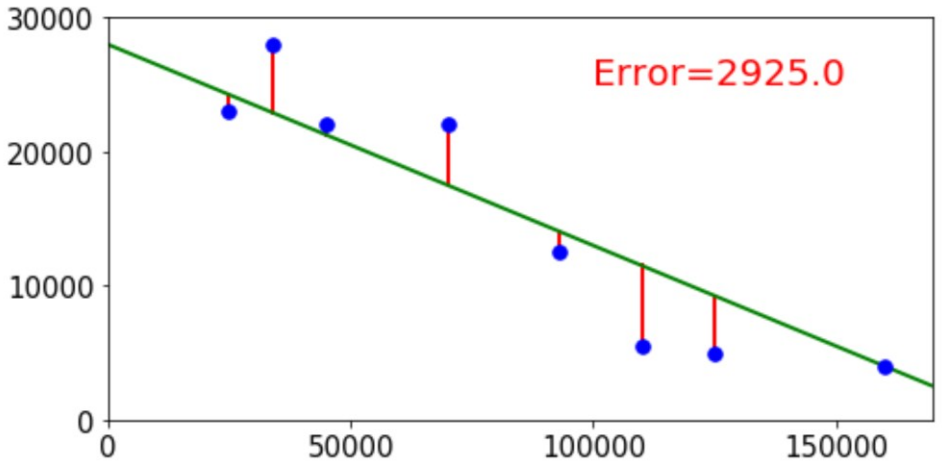
$$b \leftarrow b - \gamma \frac{\partial J}{\partial b}$$

$$\vec{\theta}^{(t+1)} \leftarrow \vec{\theta}^{(t)} - \gamma \frac{dJ}{d\vec{\theta}}$$

LINEAR REGRESSION:
LINEAR ALGEBRA FORMULATION

Linear Algebra Formulation

Constructing the design matrix



Goal: Replace summations with linear algebra

$$J(w,b) = \frac{1}{N}\sum_{i}^{N}\left(y^{(i)} - \left(b + \sum_{j}^{M} w_j x_j^{(i)}\right)\right)^2$$

$$= \frac{1}{N}\sum_{i}^{N}\left(y^{(i)} - \left(\boldsymbol{\theta}^\top \mathbf{x}^{(i)}\right)\right)^2$$ ← From previous lecture

$$= \frac{1}{N}\|\boldsymbol{y} - X\boldsymbol{\theta}\|_2^2$$

Linear Algebra Formulation

Constructing the design matrix

Reminder (Lec 2) Dataset notation

$$\mathcal{D} = \left\{\left(y^{(i)}, \mathbf{x}^{(i)}\right)\right\}_{i=1}^{N}$$

$$= \left\{\left(y^{(i)}, x_1^{(i)}, x_2^{(i)}, x_3^{(i)}, x_4^{(i)}\right)\right\}_{i=1}^{N}$$

Linear algebra can represent all data

$\mathbf{y} \in \{0,1,2\}^N$

$X \in \mathbb{R}^{N\times 4}$ (design matrix)

Species	Sepal Length	Sepal Width	Petal Length	Petal Width
0	4.3	3.0	1.1	0.1
0	4.9	3.6	1.4	0.1
0	5.3	3.7	1.5	0.2
1	4.9	2.4	3.3	1.0
1	5.7	2.8	4.1	1.3
1	6.3	3.3	4.7	1.6
2	5.9	3.0	5.1	1.8

Linear Algebra Formulation

$$\hat{y}^{(i)} = b + \sum_{j}^{M} w_j x_j^{(i)}$$

Constructing the design matrix

$\vec{y}$

Species	Sepal Length	Sepal Width	Petal Length	Petal Width
0	4.3	3.0	1.1	0.1
0	4.9	3.6	1.4	0.1
0	5.3	3.7	1.5	0.2
1	4.9	2.4	3.3	1.0
1	5.7	2.8	4.1	1.3
1	6.3	3.3	4.7	1.6
2	5.9	3.0	5.1	1.8

Linear Algebra Formulation

$$\hat{y}^{(i)} = b + \sum_{j}^{M} w_j x_j^{(i)}$$

Constructing the design matrix

$\vec{y}$

Species
0
0
0
1
1
1
2

Sepal Length	Sepal Width	Petal Length	Petal Width
4.3	3.0	1.1	0.1
4.9	3.6	1.4	0.1
5.3	3.7	1.5	0.2
4.9	2.4	3.3	1.0
5.7	2.8	4.1	1.3
6.3	3.3	4.7	1.6
5.9	3.0	5.1	1.8

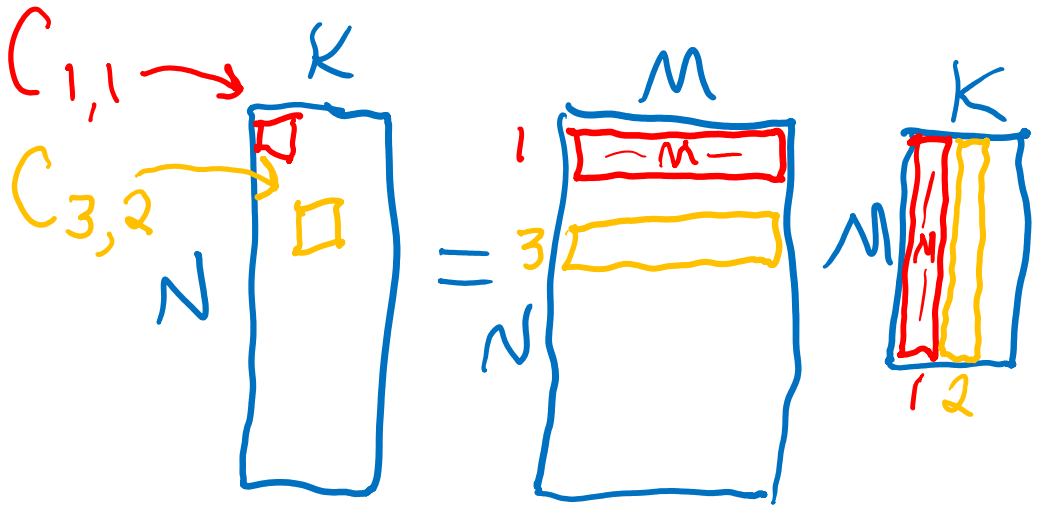
Aside: Matrix multiplication dimension check

$$C = AB$$

$$[N \times K] = [N \times M][M \times K]$$

inner dimension must be the same

outer dimensions become output dimensions



$$C_{1,1} = \sum_{j=1}^{M} A_{1j} B_{j1}$$

$$C_{3,2} = \sum_{j=1}^{M} A_{3j} B_{j2}$$

Linear Algebra Formulation

$$\hat{y}^{(i)} = b + \sum_{j}^{M} w_j x_j^{(i)}$$

$$\hat{y}^{(i)} = \boldsymbol{\theta}^\top \mathbf{x}^{(i)}$$

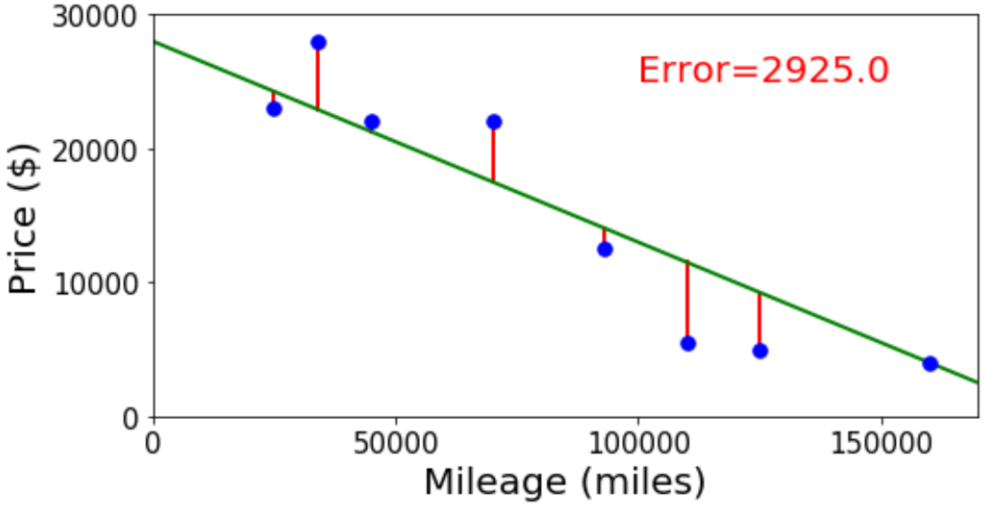
Constructing the design matrix

$\vec{y}$

Species	(account for bias)	Sepal Length	Sepal Width	Petal Length	Petal Width
0	1	4.3	3.0	1.1	0.1
0	1	4.9	3.6	1.4	0.1
0	1	5.3	3.7	1.5	0.2
1	1	4.9	2.4	3.3	1.0
1	1	5.7	2.8	4.1	1.3
1	1	6.3	3.3	4.7	1.6
2	1	5.9	3.0	5.1	1.8

Poll 3

Now that we have $\mathbf{y} = X\boldsymbol{\theta}$, we can solve this system of equations to get the best $\boldsymbol{\theta}$?



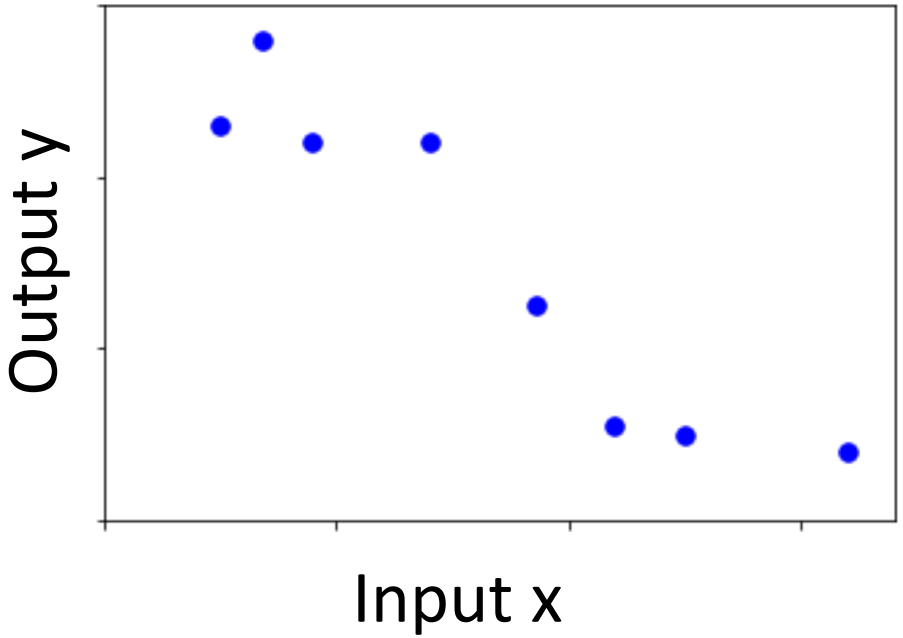
A.

B.

C.

Linear Alegebra Formulation

Solving a system of equations? Objective function value, $J(\theta)$?



LINEAR REGRESSION: CLOSED-FORM SOLUTION

Super quick with Matrix Calculus (which you probably haven't learned)

Solving for $\boldsymbol{\theta}$ by finding the critical point where $\nabla J(\theta) = \mathbf{0}$

(Expert level: The super speedy version, that has too many tricks)

$$J(\boldsymbol{\theta}) = \frac{1}{N}\|\mathbf{y} - X\boldsymbol{\theta}\|_2^2$$

Repeat with more accessible linear algebra tricks

Expanding objective before computing gradient

$$J(\boldsymbol{\theta}) = \frac{1}{N}\|\mathbf{y} - X\boldsymbol{\theta}\|_2^2$$

$$= \frac{1}{N}\left(\mathbf{y}^T\mathbf{y} - 2\boldsymbol{\theta}^T X^T\mathbf{y} + \boldsymbol{\theta}^T X^T X\boldsymbol{\theta}\right)$$

From a linear algebra cheatsheat

$$\|\mathbf{z}\|_2^2 = \mathbf{z}^\top\mathbf{z}$$

$$(\mathbf{z} - \mathbf{u})^\top = \mathbf{z}^\top - \boldsymbol{u}^\top$$

$$(A\mathbf{v})^\top = \mathbf{v}^\top A^\top$$

$$(A - B)(C - D)$$
$$= AC - BC - AD + BD$$

$$\mathbf{v}^\top A\boldsymbol{z} = (\mathbf{z}^\top A^\top\mathbf{v})^\top$$

$\mathbf{v}^\top A\boldsymbol{z}$ is a scalar, c

$$c^\top = c$$

Repeat with more accessible linear algebra tricks

Expanding objective before computing gradient

$$
\begin{aligned}
J(\boldsymbol{\theta}) &= \frac{1}{N}\|\mathbf{y} - X\boldsymbol{\theta}\|_2^2 \\
&= \frac{1}{N}(\mathbf{y} - X\boldsymbol{\theta})^T(\mathbf{y} - X\boldsymbol{\theta}) \\
&= \frac{1}{N}\left(\mathbf{y}^T - \boldsymbol{\theta}^T X^T\right)(\mathbf{y} - X\boldsymbol{\theta}) \\
&= \frac{1}{N}\left(\mathbf{y}^T\mathbf{y} - \boldsymbol{\theta}^T X^T\mathbf{y} - \mathbf{y}^T X\boldsymbol{\theta} + \boldsymbol{\theta}^T X^T X\boldsymbol{\theta}\right) \\
&= \frac{1}{N}\left(\mathbf{y}^T\mathbf{y} - 2\boldsymbol{\theta}^T X^T\mathbf{y} + \boldsymbol{\theta}^T X^T X\boldsymbol{\theta}\right)
\end{aligned}
$$

From a linear algebra cheatsheat

$$\|\mathbf{z}\|_2^2 = \mathbf{z}^\top\mathbf{z}$$

$$(\mathbf{z} - \mathbf{u})^\top = \mathbf{z}^\top - \boldsymbol{u}^\top$$

$$(A\mathbf{v})^\top = \mathbf{v}^\top A^\top$$

$$(A - B)(C - D) = AC - BC - AD + BD$$

$$\mathbf{v}^\top A\boldsymbol{z} = (\mathbf{z}^\top A^\top\mathbf{v})^\top$$

$\mathbf{v}^\top A\boldsymbol{z}$ is a scalar, c

$$c^\top = c$$

Linear Regression

Gradient of objective with respect to parameters

$$J(\boldsymbol{\theta}) = \frac{1}{N}\|\boldsymbol{y} - X\boldsymbol{\theta}\|_2^2$$

$$= \frac{1}{N}\left(\boldsymbol{y}^T\boldsymbol{y} - 2\boldsymbol{\theta}^T X^T \boldsymbol{y} + \boldsymbol{\theta}^T X^T X \boldsymbol{\theta}\right)$$

$$\nabla J(\boldsymbol{\theta}) = \frac{1}{N}$$

From a matrix calculus cheatsheat

$$\frac{\partial \boldsymbol{z}^T \boldsymbol{u}}{\partial \boldsymbol{z}} = \boldsymbol{u}$$

-- or --

$$\frac{\partial \boldsymbol{z}^T \boldsymbol{u}}{\partial \boldsymbol{z}} = \boldsymbol{u}^T$$

$$\frac{\partial \boldsymbol{z}^T A \boldsymbol{z}}{\partial \boldsymbol{z}} = \left(A + A^T\right)\boldsymbol{z}$$

-- or --

$$\frac{\partial \boldsymbol{z}^T A \boldsymbol{z}}{\partial \boldsymbol{z}} = \boldsymbol{z}^T\left(A + A^T\right)$$

Linear Regression

Gradient of objective with respect to parameters

$$J(\boldsymbol{\theta}) = \frac{1}{N}\|\boldsymbol{y} - X\boldsymbol{\theta}\|_2^2$$

$$= \frac{1}{N}\left(\boldsymbol{y}^T\boldsymbol{y} - 2\boldsymbol{\theta}^T X^T \boldsymbol{y} + \boldsymbol{\theta}^T X^T X \boldsymbol{\theta}\right)$$

$$\nabla J(\boldsymbol{\theta}) = \frac{1}{N}\left(0 - 2X^T\boldsymbol{y} + 2\boldsymbol{\theta}^T X^T X\right)$$

From a matrix calculus cheatsheat

$$\frac{\partial \boldsymbol{z}^T\boldsymbol{u}}{\partial \boldsymbol{z}} = \boldsymbol{u}$$

-- or --

$$\frac{\partial \boldsymbol{z}^T\boldsymbol{u}}{\partial \boldsymbol{z}} = \boldsymbol{u}^T$$

$$\frac{\partial \boldsymbol{z}^T A\boldsymbol{z}}{\partial \boldsymbol{z}} = \left(A + A^T\right)\boldsymbol{z}$$

-- or --

$$\frac{\partial \boldsymbol{z}^T A\boldsymbol{z}}{\partial \boldsymbol{z}} = \boldsymbol{z}^T\left(A + A^T\right)$$

Linear Regression

Gradient of objective with respect to parameters

$$J(\boldsymbol{\theta}) = \frac{1}{N}\|\boldsymbol{y} - X\boldsymbol{\theta}\|_2^2$$

$$= \frac{1}{N}\left(\boldsymbol{y}^T\boldsymbol{y} - 2\boldsymbol{\theta}^T X^T \boldsymbol{y} + \boldsymbol{\theta}^T X^T X\boldsymbol{\theta}\right)$$

$$\nabla J(\boldsymbol{\theta}) = \frac{1}{N}\left(0 - 2X^T\boldsymbol{y} + \cancel{2\boldsymbol{\theta}^T X^T X}\right)$$

$$= \frac{1}{N}\left(0 - 2X^T\boldsymbol{y} + 2X^T X\boldsymbol{\theta}\right)$$

From a matrix calculus cheatsheat

$$\frac{\partial \boldsymbol{z}^T\boldsymbol{u}}{\partial \boldsymbol{z}} = \boldsymbol{u}$$

-- or --

$$\frac{\partial \boldsymbol{z}^T\boldsymbol{u}}{\partial \boldsymbol{z}} = \boldsymbol{u}^T$$

$$\frac{\partial \boldsymbol{z}^T A\boldsymbol{z}}{\partial \boldsymbol{z}} = \left(A + A^T\right)\boldsymbol{z}$$

-- or --

$$\frac{\partial \boldsymbol{z}^T A\boldsymbol{z}}{\partial \boldsymbol{z}} = \boldsymbol{z}^T\left(A + A^T\right)$$

Linear Regression

Gradient of objective with respect to parameters

$$J(\boldsymbol{\theta}) = \frac{1}{N}\|\boldsymbol{y} - X\boldsymbol{\theta}\|_2^2$$

$$= \frac{1}{N}\left(\boldsymbol{y}^T\boldsymbol{y} - 2\boldsymbol{\theta}^T X^T \boldsymbol{y} + \boldsymbol{\theta}^T X^T X\boldsymbol{\theta}\right)$$

$$\nabla J(\boldsymbol{\theta}) = \frac{1}{N}\left(0 - 2X^T\boldsymbol{y} + \cancel{2\boldsymbol{\theta}^T X^T X}\right)$$

$$= \frac{1}{N}\left(0 - 2X^T\boldsymbol{y} + 2X^T X\boldsymbol{\theta}\right)$$

$$= \frac{2}{N}\left(-X^T\boldsymbol{y} + X^T X\boldsymbol{\theta}\right)$$

From a matrix calculus cheatsheat

$$\frac{\partial \boldsymbol{z}^T\boldsymbol{u}}{\partial \boldsymbol{z}} = \boldsymbol{u}$$

-- or --

$$\frac{\partial \boldsymbol{z}^T\boldsymbol{u}}{\partial \boldsymbol{z}} = \boldsymbol{u}^T$$

$$\frac{\partial \boldsymbol{z}^T A\boldsymbol{z}}{\partial \boldsymbol{z}} = \left(A + A^T\right)\boldsymbol{z}$$

-- or --

$$\frac{\partial \boldsymbol{z}^T A\boldsymbol{z}}{\partial \boldsymbol{z}} = \boldsymbol{z}^T\left(A + A^T\right)$$

Linear Regression

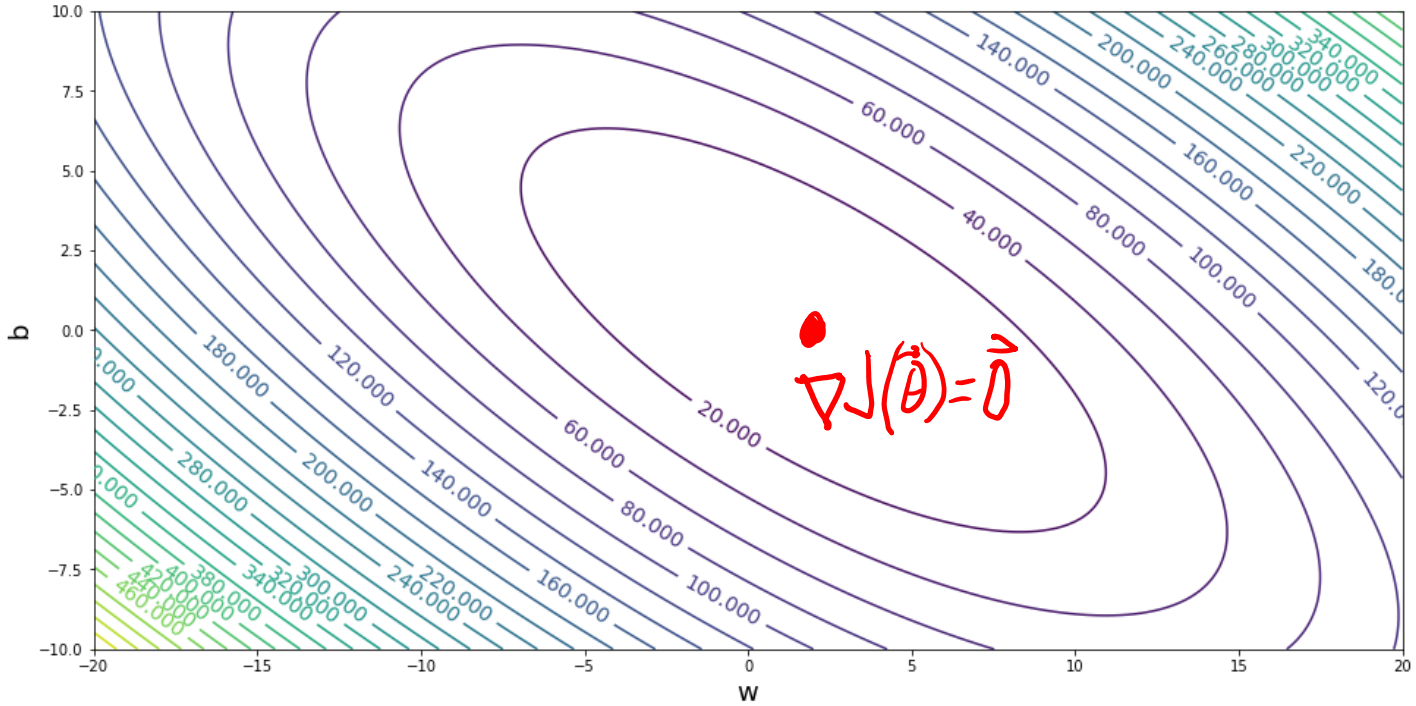
Closed-form solution

$$\nabla J(\boldsymbol{\theta}) = \frac{2}{N}\left(-X^T\boldsymbol{y} + X^TX\boldsymbol{\theta}\right)$$

$$\frac{2}{N}\left(-X^T\boldsymbol{y} + X^TX\boldsymbol{\theta}\right) = \mathbf{0}$$

$$X^TX\boldsymbol{\theta} = X^T\boldsymbol{y}$$ ← The famous normal equations

$$\widehat{\boldsymbol{\theta}} = \left(X^TX\right)^{-1}X^T\boldsymbol{y}$$ ← Solve normal equations to get estimate for the linear regression parameters, $\widehat{\boldsymbol{\theta}}$



CALCULUS TIMEOUT

Calculus with Linear Algebra

Collection of partial derivatives for **every pair of scalar inputs and outputs**

Vector in, scalar out

Gradient

$y = f(\mathbf{x}) \qquad y \in \mathbb{R}, \;\; \mathbf{x} \in \mathbb{R}^M$

Different ways to organize this collection:

$$\frac{\partial f}{\partial \mathbf{x}} = \begin{bmatrix} \frac{\partial y}{\partial x_1} & \cdots & \frac{\partial y}{\partial x_M} \end{bmatrix}$$

-- or --

$$\frac{\partial f}{\partial \mathbf{x}} = \begin{bmatrix} \frac{\partial y}{\partial x_1} \\ \vdots \\ \frac{\partial y}{\partial x_M} \end{bmatrix}$$

Calculus with Linear Algebra

Functions with different combinations of input and output sizes

$$y = \mathrm{f}(x)$$

$$y = \mathrm{f}\left(\begin{bmatrix} x_1 \\ \vdots \\ x_M \end{bmatrix}\right)$$

$$y = \mathrm{f}\left(\begin{bmatrix} X_{1,1} & \cdots & X_{1,L} \\ \vdots & \ddots & \vdots \\ X_{M,1} & \cdots & X_{M,L} \end{bmatrix}\right)$$

$$\begin{bmatrix} y_1 \\ \vdots \\ y_N \end{bmatrix} = \mathrm{f}(x)$$

$$\begin{bmatrix} y_1 \\ \vdots \\ y_N \end{bmatrix} = \mathrm{f}\left(\begin{bmatrix} x_1 \\ \vdots \\ x_M \end{bmatrix}\right)$$

$$\begin{bmatrix} y_1 \\ \vdots \\ y_N \end{bmatrix} = \mathrm{f}\left(\begin{bmatrix} X_{1,1} & \cdots & X_{1,L} \\ \vdots & \ddots & \vdots \\ X_{M,1} & \cdots & X_{M,L} \end{bmatrix}\right)$$

$$\begin{bmatrix} Y_{1,1} & \cdots & Y_{1,K} \\ \vdots & \ddots & \vdots \\ Y_{N,K} & \cdots & Y_{N,K} \end{bmatrix} = \mathrm{f}(x)$$

$$\begin{bmatrix} Y_{1,1} & \cdots & Y_{1,K} \\ \vdots & \ddots & \vdots \\ Y_{N,K} & \cdots & Y_{N,K} \end{bmatrix} = \mathrm{f}\left(\begin{bmatrix} x_1 \\ \vdots \\ x_M \end{bmatrix}\right)$$

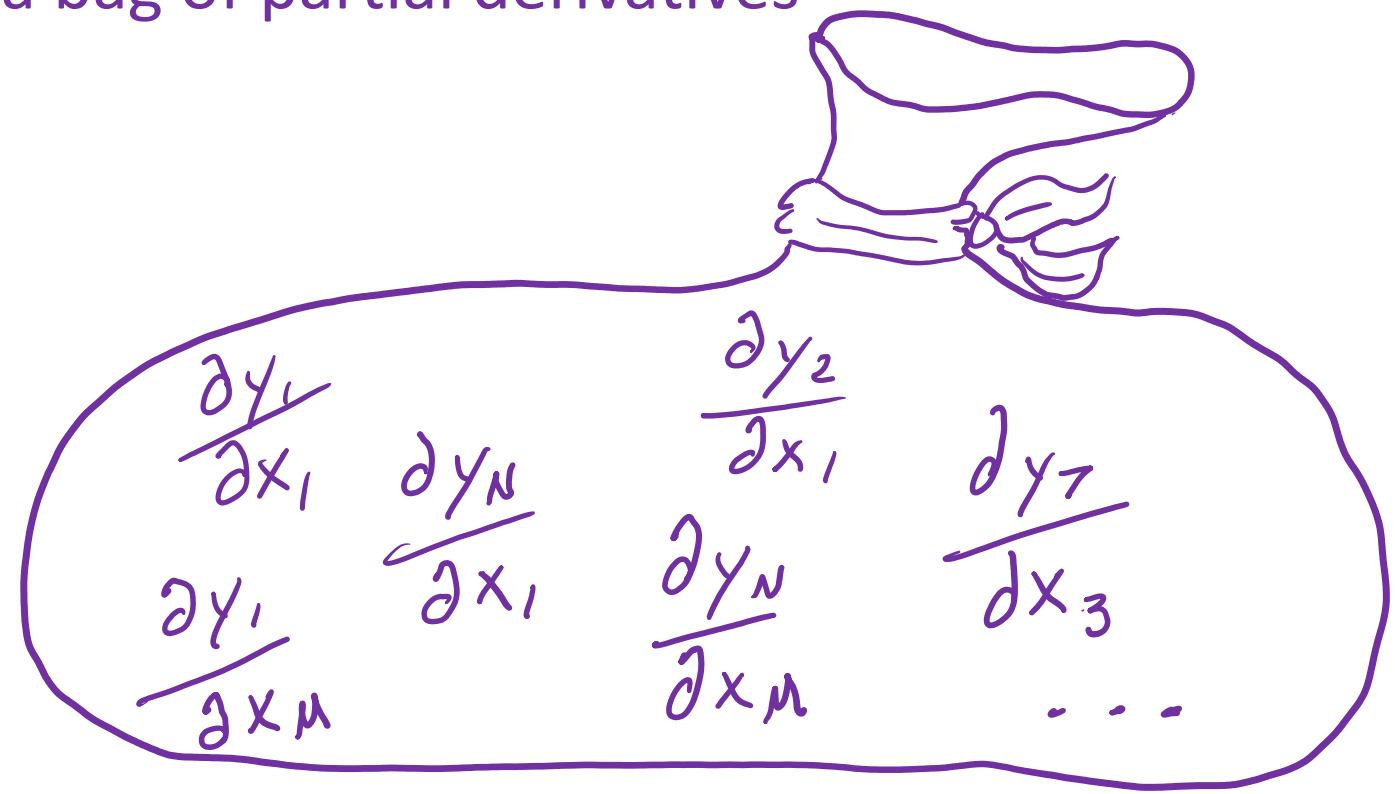
$$\begin{bmatrix} Y_{1,1} & \cdots & Y_{1,K} \\ \vdots & \ddots & \vdots \\ Y_{N,K} & \cdots & Y_{N,K} \end{bmatrix} = \mathrm{f}\left(\begin{bmatrix} X_{1,1} & \cdots & X_{1,L} \\ \vdots & \ddots & \vdots \\ X_{M,1} & \cdots & X_{M,L} \end{bmatrix}\right)$$

Calculus with Linear Algebra

Collection of partial derivatives for **every pair of scalar inputs and outputs**

One way to think of it: a bag of partial derivatives

$$\begin{bmatrix} y_1 \\ \vdots \\ y_N \end{bmatrix} = \mathrm{f}\left(\begin{bmatrix} x_1 \\ \vdots \\ x_M \end{bmatrix}\right)$$

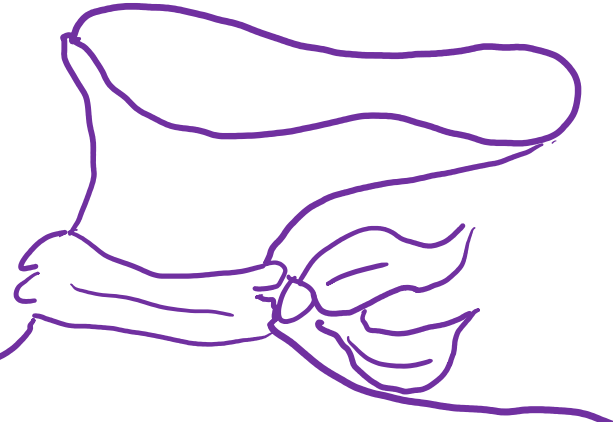


Calculus with Linear Algebra

Collection of partial derivatives for **every pair of scalar inputs and outputs**

One way to think of it: a bag of partial derivatives

$$\begin{bmatrix} Y_{1,1} & \cdots & Y_{1,K} \\ \vdots & \ddots & \vdots \\ Y_{N,1} & \cdots & Y_{N,K} \end{bmatrix} = \mathrm{f}\left(\begin{bmatrix} X_{1,1} & \cdots & X_{1,L} \\ \vdots & \ddots & \vdots \\ X_{M,1} & \cdots & X_{M,L} \end{bmatrix}\right)$$



$$\frac{\partial Y_{11}}{\partial X_{11}} \quad \frac{\partial Y_{1K}}{\partial X_{11}} \quad \frac{\partial Y_{NK}}{\partial X_{ML}} \quad \frac{\partial Y_{1K}}{\partial X_{ML}}$$

$$\frac{\partial Y_{NK}}{\partial X_{11}} \quad \frac{\partial Y_{2,3}}{\partial X_{1,5}} \quad \frac{\partial Y_{11}}{\partial X_{1L}} \quad \cdots$$

Organization Convention: Numerator Layout

$y = f(x)$	$x \in \mathbb{R}$	$\mathbf{x} \in \mathbb{R}^{\mathrm{M}}$	$X \in \mathbb{R}^{\mathrm{M} \times L}$
$y \in \mathbb{R}$	$\frac{\partial y}{\partial x} \in \mathbb{R}$	$\frac{\partial y}{\partial \mathbf{x}} \in \mathbb{R}^{1 \times M}$ Output by input	$\frac{\partial y}{\partial X} \in \mathbb{R}^{L \times M}$ Transpose of input
$\mathbf{y} \in \mathbb{R}^{N}$	$\frac{\partial \mathbf{y}}{\partial x} \in \mathbb{R}^{N \times 1}$ Output by input	$\frac{\partial \mathbf{y}}{\partial \mathbf{x}} \in \mathbb{R}^{N \times M}$ Output by input	$\frac{\partial \mathbf{y}}{\partial X} \in \mathbb{R}^{?}$?
$Y \in \mathbb{R}^{N \times K}$	$\frac{\partial Y}{\partial x} \in \mathbb{R}^{N \times K}$ Size of output	$\frac{\partial Y}{\partial \mathbf{x}} \in \mathbb{R}^{?}$?	$\frac{\partial Y}{\partial X} \in \mathbb{R}^{?}$?

Organization Convention: Denominator Layout

$y = f(x)$	$x \in \mathbb{R}$	$\mathbf{x} \in \mathbb{R}^{\mathrm{M}}$	$X \in \mathbb{R}^{\mathrm{M}\times L}$
$y \in \mathbb{R}$	$\frac{\partial y}{\partial x} \in \mathbb{R}$	$\frac{\partial y}{\partial \mathbf{x}} \in \mathbb{R}^{M\times 1}$ Input by output	$\frac{\partial y}{\partial X} \in \mathbb{R}^{M\times L}$ Size of input
$\mathbf{y} \in \mathbb{R}^{N}$	$\frac{\partial \mathbf{y}}{\partial x} \in \mathbb{R}^{1\times N}$ Input by output	$\frac{\partial \mathbf{y}}{\partial \mathbf{x}} \in \mathbb{R}^{M\times N}$ Input by output	$\frac{\partial \mathbf{y}}{\partial X} \in \mathbb{R}^{?}$?
$Y \in \mathbb{R}^{N\times K}$	$\frac{\partial Y}{\partial x} \in \mathbb{R}^{K\times N}$ Transpose of output	$\frac{\partial Y}{\partial \mathbf{x}} \in \mathbb{R}^{?}$?	$\frac{\partial Y}{\partial X} \in \mathbb{R}^{?}$?

Cheatsheet: Derivatives with Respect to Vectors

	Numerator layout	Denominator layout	Notes
$\frac{\partial}{\partial \mathbf{v}}\ \mathbf{v}$	I_N	I_N	$\mathbf{v} \in \mathbb{R}^N$
$\frac{\partial}{\partial \mathbf{v}}\ \mathbf{v}^T$	I_N	I_N	$\mathbf{v} \in \mathbb{R}^N$
$\frac{\partial}{\partial \mathbf{v}}\ t\mathbf{v}$	tI_N	tI_N	$\mathbf{v} \in \mathbb{R}^N$
$\frac{\partial}{\partial \mathbf{u}}\ \mathbf{u}^T\mathbf{v}$	$\mathbf{v}^T$	$\mathbf{v}$	
$\frac{\partial}{\partial \mathbf{v}}\ \mathbf{u}^T\mathbf{v}$	$\mathbf{u}^T$	$\mathbf{u}$	
$\frac{\partial}{\partial \mathbf{v}}\ \mathbf{v}^T\mathbf{v}$	$2\mathbf{v}^T$	$2\mathbf{v}$	
$\frac{\partial}{\partial \mathbf{v}}\ \mathbf{v}^T A\mathbf{v}$	$\mathbf{v}^T(A + A^T)$	$(A + A^T)\mathbf{v}$	
$\frac{\partial}{\partial \mathbf{v}}\ \mathbf{v}^T A\mathbf{v}$	$2\mathbf{v}^T A$	$2A\mathbf{v}$	If $A = A^T$
$\frac{\partial}{\partial \mathbf{v}}\ A\mathbf{v}$	A	A^T	
$\frac{\partial}{\partial \mathbf{v}}\ \mathbf{v}^T A$	A^T	A	